

Pedro Beber de Queiroz Vidal

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LLM agents and computer vision researcher, designing multi-agent architectures at Unicamp's HIAAC as part of the HIAAC project, with a PROEX/Unicamp graduate scholarship. Eleven publications in four years, three as first author, and podium finishes in four international challenges, including **1st place** in a sub-task of the FRCSyn Challenge at WACV 2024.

EDUCATION

University of Campinas (Unicamp)

MSc in Computer Science — PROEX/Unicamp scholarship holder

Thesis: architectures for LLM multi-agent systems, integrating dynamic memory retrieval and inter-agent communication.

Campinas, SP, Brazil

2026 — Mar 2028 (expected)

Federal University of Paraná (UFPR)

BSc in Computer Science

Thesis on synthetic facial data, extended into a survey submitted to IEEE Access.

Curitiba, PR, Brazil

2020 — Jul 2025

Federal Institute of Paraná (IFPR)

Technical Diploma in Informatics

Irati, PR, Brazil

2016 — 2019

RESEARCH EXPERIENCE

HIAAC, University of Campinas (Unicamp)

Graduate Researcher, LLM Multi-Agent Systems

Campinas, SP, Brazil

Aug 2026 — Present

- Design a multi-agent architecture for LLM agents that couples dynamic memory retrieval with an optimized inter-agent communication protocol, evaluated against state-of-the-art agentic benchmarks.
- Benchmark reasoning and orchestration strategies — ReAct, chain-of-thought, in-context learning, task decomposition — to isolate which choices actually move task success under controlled conditions.
- Authored a standalone literature review of LLM multi-agent systems, mapping communication topologies, memory mechanisms, and evaluation practice across the field.
- Responsible for the LLM-agent systems component of the HIAAC project.

BOVIFOCR, UFPR-UNICO Partnership

Undergraduate Researcher, Computer Vision

Curitiba, PR, Brazil

Oct 2021 — Mar 2025

- Built the submission that took **1st place** in Sub-Task 2.1 of the FRCSyn Challenge at WACV 2024, and the SDFR entries that placed 3rd in both tasks at FG 2024.
- Trained and evaluated face recognition models on synthetic identity datasets of millions of images, adapting InsightFace and running the pipeline end to end from data processing to benchmark reporting.
- Quantified the impact of synthetic training data across eight standard benchmarks, identifying which dataset characteristics predict recognition performance; published at SIBGRAPI 2024 and extended into a survey submitted to IEEE Access.
- Authored the group's masked face recognition benchmark (SIBGRAPI 2022) and represented it at the Occluded Face Recognition competition at IJCB 2022.
- Built reproducible benchmark protocols over standardized datasets to separate genuine model gains from training-data effects, tracking every run in Weights & Biases.

- Collected liveness samples in the field, assisting elderly participants unfamiliar with the technology, and drove adoption of the collection app on social networks.

University of Louisiana at Lafayette

Visiting Researcher, Exchange Program

Lafayette, LA, USA

Mar — Jun 2023

- Improved occluded face verification by extending Earth Mover's Distance as the matching metric, validating the gain through False Non-Match Rate at fixed False Match Rate; published at SIBGRAPI 2023.
- Selected for a summer school on innovation and startups with international students, followed by an internship at the university's computer science research institute.
- Proposed a financial planning application for university students covering tuition, living costs, and scholarship tracking.

HONORS AND COMPETITIONS

FRCSyn Challenge, 1st Edition, WACV 2024 — **1st place**, Sub-Task 2.1; 3rd, 4th and 4th in Sub-Tasks 1.1, 1.2 and 2.2. 2023

SDFR Competition, FG 2024 — 3rd place in both tasks of Synthetic Data for Face Recognition. 2024 — 2025

FRCSyn Challenge, 2nd Edition, CVPR 2024 — 6th place, Sub-Task 2.1. 2024

PROEX/Unicamp graduate scholarship — graduate research funding. 2026

OBMEP, Brazilian Mathematics Olympiad — Honorable Mention. 2019

PROJECTS

Federated Learning for Privacy-Preserving Facial Recognition — [code](#) 2026 · Unicamp

2D face recognition trained without centralizing biometric data, combining federated clients with synthetic facial data to hold accuracy under non-IID splits, limited client compute, and scarce local data.

Comparative Evaluation of Synthetic Facial Datasets — [thesis](#) 2025 · UFPR

Trained face recognition models on multiple synthetic datasets and scored them across eight benchmarks, isolating the dataset characteristics and quality trade-offs behind stronger and weaker performance.

Training Pipeline for Synthetic-Data Competitions — [code](#) 2024 · UFPR

The pipeline behind the SDFR and FRCSyn submissions: InsightFace adapted for training and evaluation on million-image synthetic identity datasets, with results organized for comparison across datasets, model configurations, and benchmark protocols.

Robotics Team, Irathunder Group 2017 — 2019 · IFPR

Programmed and built LEGO robots for rescue, sumo, and line-following events; competed in two editions of the Brazilian Robotics Olympiad (OBR).

TECHNICAL SKILLS

LLMs, agents Multi-agent systems, inter-agent communication, dynamic memory retrieval, ReAct, chain-of-thought, in-context learning, task decomposition, agentic benchmarking.

Machine learning PyTorch, TensorFlow, MXNet, Keras, OpenCV, InsightFace, Flower; generative models, face recognition, liveness detection, knowledge distillation, LoRA and parameter-efficient fine-tuning, Federated Learning.

Data NumPy, Pandas, Scikit-learn, Matplotlib; dataset validation, data cleaning, exploratory analysis, metric computation, statistical comparison of results.

Engineering	Python (advanced), Git, Linux, shell scripting, Conda, Jupyter, Weights & Biases; reproducible experiment design, distributed training, large-scale dataset processing.
Languages	Portuguese (native), English (fluent).

PUBLICATIONS

FIRST AUTHOR

1. **Vidal, Pedro**, Bernardo Biesseck, et al. (2024). "A Comparative Study on Synthetic Facial Data Generation Techniques for Face Recognition". In: *2024 37th SIBGRAPI Conference on Graphics, Patterns and Images*. sol.sbc.org.br
2. **Vidal, Pedro**, Henry Chu, et al. (2023). "Extending Earth Mover's Distance to Occluded Face Verification". In: *2023 36th SIBGRAPI Conference on Graphics, Patterns and Images*. IEEE, pp. 49–54. sibgrapi.sid.inpe.br
3. **Vidal, Pedro**, Roger Leitzke Granada, et al. (2022). "A Benchmark on Masked Face Recognition". In: *Proceedings... Conference on Graphics, Patterns and Images, 35 (SIBGRAPI)*. urlib.net
4. **Vidal, Pedro Beber** (2024). *A Comparative Study on Synthetic Facial Data Generation Techniques for Face Recognition*. Undergraduate thesis, UFPR, Curitiba. bit.ly/3V03Efp

CO-AUTHORED

5. Melzi, Pietro et al. (2024). "FRCSyn-onGoing: Benchmarking and comprehensive evaluation of real and synthetic data to improve face recognition systems". In: *Information Fusion* 107, 102322. doi.org
6. DeAndres-Tame, Ivan et al. (2025). "Second FRCSyn-onGoing: Winning solutions and post-challenge analysis to improve face recognition with synthetic data". In: *Information Fusion* 120, 103099. doi.org
7. Biesseck, Bernardo, **Pedro Vidal**, et al. (2024). "TCDiff: Triple Condition Diffusion Model with 3D Constraints for Stylizing Synthetic Faces". In: *2024 37th SIBGRAPI Conference on Graphics, Patterns and Images*, pp. 1–6. scholar.google.com
8. Biesseck, Bernardo, **Pedro Vidal**, et al. (2023). "3D Face Reconstruction as Complementary Data to Enhance Face Recognition". In: *2023 36th SIBGRAPI Conference on Graphics, Patterns and Images*, pp. 61–66. scholar.google.com
9. Shahreza, Hatef Otroshi et al. (2024). "SDFR: Synthetic Data for Face Recognition Competition". In: *2024 IEEE 18th International Conference on Automatic Face and Gesture Recognition (FG)*. IEEE, pp. 1–9. ieeexplore.ieee.org
10. DeAndres-Tame, Ivan et al. (2024). "FRCSyn Challenge at CVPR 2024: Face Recognition Challenge in the Era of Synthetic Data". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3173–3183. openaccess.thecvf.com
11. Melzi, Pietro et al. (2024). "FRCSyn Challenge at WACV 2024: Face Recognition Challenge in the Era of Synthetic Data". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 892–901. openaccess.thecvf.com
12. Neto, Pedro C et al. (2022). "OCFR 2022: Competition on Occluded Face Recognition from Synthetically Generated Structure-Aware Occlusions". In: *International Joint Conference on Biometrics 2022*. ieeexplore.ieee.org